

3.3. UNSMOOTHING RETURNS

To account for the infrequent trading bias, we need to go from Panel A of Figure 13.2, which samples quarterly, to Figure B, which samples daily. That is, the quarterly observed returns are too smooth, and we need to tease out the true, noisier returns. This process is called *unsmoothing* or *de-smoothing*, and the first algorithms to do this were developed by David Geltner (1991), a noted professor of real estate at MIT, and Stephen Ross and Randall Zisler (1991). Ross is the same professor who developed multifactor models (see chapter 6) and Zisler is a real estate professional who started his career as an academic. Ross and Zisler's work originally grew out of a series of reports written for Goldman Sachs in the late 1980s. This methodology has been extended in what is now an extensive literature.

Unsmoothing is a *filtering problem*. Filtering algorithms are normally used to separate signals from noise. When we're driving on a freeway and talking on a cell phone, our phone call encounters interference—from highway overpasses and tall buildings—or the reception becomes patchy when we pass through an area without enough cell phone towers. Telecommunication engineers use clever algorithms to enhance the signal, which carries our voice, against all the static. The full transmission contains both the signal and noise, and so the true signal is less volatile than the full transmission. Thus standard filtering problems are designed to remove noise. The key difference is that unsmoothing *adds* noise back to the reported returns to uncover the true returns.

To illustrate the Geltner-Ross-Zisler unsmoothing process, denote the true return at the end of period t as r_t , which is unobservable, and the reported return as r_t^* , which is observable. Suppose the observable returns follow

$$r_t^* = c + \phi r_{t-1}^* + \varepsilon_t, \quad (13.1)$$

where ϕ is the autocorrelation parameter and is less than one in absolute value. Equation (13.1) is an AR(1) process, where "AR" stands for autoregressive and the "1" denotes that it captures autocorrelation effects for one lag. Assuming the observed returns are functions of current and lagged true returns (this is called a "*transfer function*" or an "*observation equation*" in the parlance of engineers), we can use equation (13.1) to invert out the true returns. If the smoothing process only involves averaging returns for this period and the past period, then we can filter the observed returns to estimate the true returns, r_t , from observed returns, r_t^* , using:

$$r_t = \frac{1}{1-\phi} r_t^* - \frac{\phi}{1-\phi} r_{t-1}^*. \quad (13.2)$$

Equation (13.2) unsmooths the observed returns. If our assumption on the transfer function is right, the observed returns implied by equation (13.2) should have zero autocorrelation. Thus, the filter takes an autocorrelated series of